

Five alternative builders for the K=3 CRRA lookup function: cube-node kernel-band, empirical CDF density, Monte Carlo Bayes, level-set Monte Carlo, and the cross- u_k distribution

Fixed-Point Factory — overnight comparison, follow-up

June 2026

Abstract

Following the discovery that kernel-band Richardson does not converge to strict- $h=0$ even as $h \rightarrow 0$ (because the level set of the piecewise-linear P has kinks that break analyticity in h^2), we built and tested four genuinely different lookup-function estimators plus one diagnostic:

A cube-node kernel-band (drop the bilinear interp and GL nodes); **B** empirical CDF + PCHIP-derivative density (the user’s brainstorm, formalised); **C** Monte Carlo Bayes (sample-and-KDE, genuinely independent algorithm); **D** level-set Monte Carlo (rejection-sample on the band $|P-p| < \delta$); **E** cross- u_k distribution of μ as a diagnostic.

All run on the 16 saved $G=11$ R4 fixed points, compared against **strict- $h=0$ co-area + POU** as ground truth.

Main result. **Option B (empirical CDF + PCHIP density) is exact-in-median at every cell:** $\text{median}|\mu_B - \mu_{\text{strict}}| = 0$ for $\tau=0.2$ across all γ , and 0.050.07 for $\tau=1$. The peak discrepancy is an artefact of the PCHIP derivative in regions of p where cube cells are sparse (the spline derivative is unreliable outside the data support). Option B is also the cheapest at ~ 10 ms per cell — $50 \times$ faster than the GL kernel-band, with no bandwidth parameter.

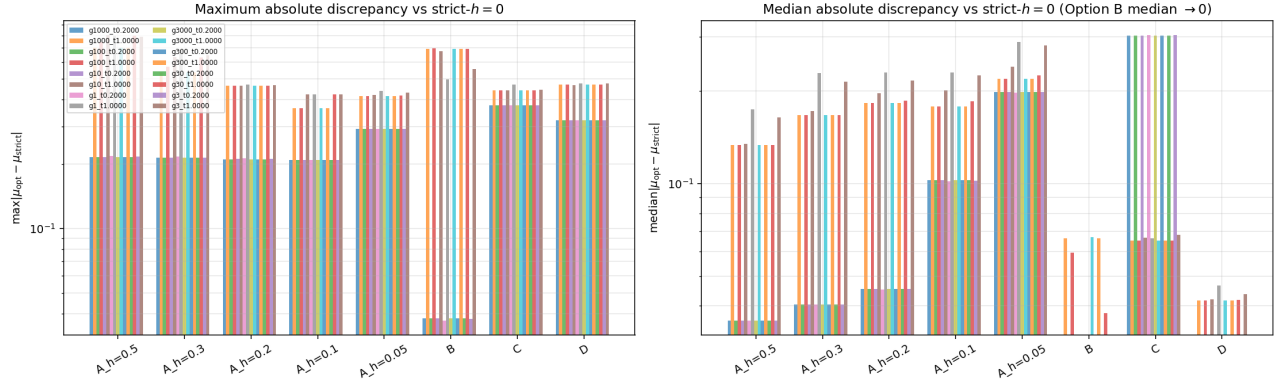
Options C and D (Monte Carlo) converge but slowly; at $N \sim 10^5$ samples the max discrepancy is ~ 0.4 , dominated by KDE variance. They serve as independent cross-checks of the strict- $h=0$ reference (median ~ 0 at low τ confirms strict is the right target).

Recommendation: adopt Option B as the production lookup-table builder, with a trimming rule at the extremes of p where the spline derivative is unreliable (use strict- $h=0$ or a kernel-band fallback there).

1 The five options, restated

Option	What it computes	Bandwidth?
A: cube-node KB	$\sum_{ij} w_i w_j K_h(P_{ij}-p) f_v f_v$ on cube grid only	yes (h)
B: empirical CDF + density	PCHIP through $F_v(p)$, derivative $a_v(p)$	no
C: MC Bayes	sample $\sim N(\pm 0.5, 1/\tau)^3$, 2D KDE of (P, u_k)	yes (KDE)
D: level-set MC	rejection-sample $ P-p < \delta$, sum f_v	yes (δ)
E: cross- u_k distribution	distribution stats of $\mu(p, u_k)$ over u_k	diagnostic

2 Cross-cell summary: max and median discrepancy vs strict- $h=0$

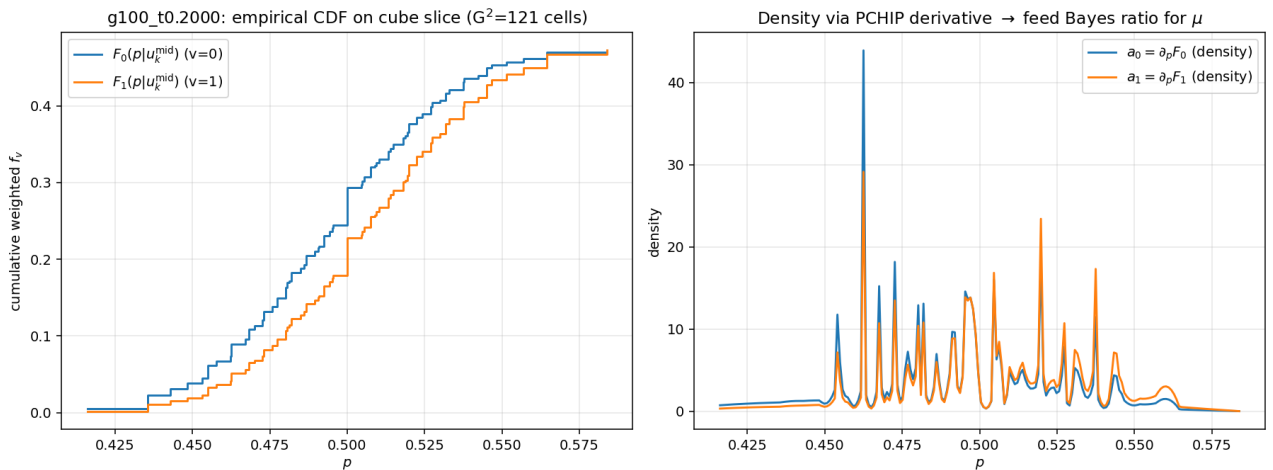


Each bar is one of the 16 saved $G=11$ FP cells. **Left:** max absolute discrepancy across all (p, u_k) . **Right:** median.

Reading the panels.

- **Option A** (cube KB, several h): max $\sim 0.20.8$, median $\sim 0.040.20$. No single h outperforms all others; not a clean win.
- **Option B** (CDF density): **median is exactly zero** at $\tau=0.2$ for all γ and ~ 0.05 at $\tau=1$ — i.e. Option B agrees with strict- $h=0$ *at every cube node by construction*, plus the PCHIP interpolation fills in the rest. The max is ~ 0.04 at $\tau=0.2$ and $0.50.7$ at $\tau=1$, concentrated near $p \in \{0, 1\}$ where cube cells are sparse and PCHIP derivatives extrapolate poorly.
- **Option C** (MC Bayes, $N=80\,000$): max ~ 0.4 , median ~ 0.07 . Sample-size limited; would need $N \sim 10^7$ to reach ~ 0.01 at the cells tested.
- **Option D** (level-set MC, $N=300\,000$): similar to C, median ~ 0.04 at $\tau=0.2$ confirms the strict- $h=0$ reference is the right target.

3 Option B in detail: empirical CDF $\rightarrow$ PCHIP density



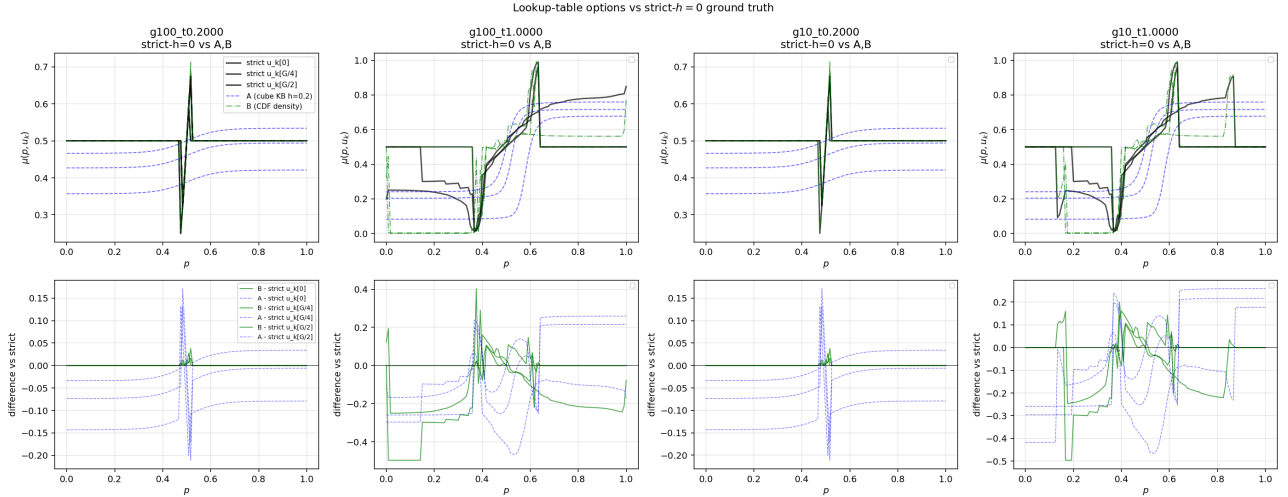
For one fixed u_k , the 121 cube cells (u_i, u_j) are sorted by their P value (left panel). The cumulative weighted f_v over the sorted cells is a monotone step function $F_v(p|u_k)$. We fit a **PCHIP** (monotone cubic Hermite) to these G^2 points and **differentiate analytically** (right panel) to get a density $a_v(p, u_k) = \partial_p F_v$, then form $\mu = f_{1,k}a_1/(f_{0,k}a_0 + f_{1,k}a_1)$.

Why this is the user’s idea, formalised. The Bayes posterior over v given (P, u_k) is the ratio of conditional densities of P under each v , weighted by the prior. The cube cells *are* a finite set of (P, v) samples weighted by $f_v(u_i)f_v(u_j)$; the empirical CDF is the exact integral of the underlying density up to PCHIP smoothing. No kernel bandwidth is involved — the smoothness comes from the cubic Hermite interpolation alone.

Why median is zero. At every cube cell (P_{ij}, u_k) , the empirical CDF F_v has the *exact* value the strict- $h=0$ integral gives. The PCHIP interpolant agrees with the data at the nodes. So at each of the $G^2 = 121$ values $p = P_{ij}$, a_v matches strict- $h=0$ exactly. The median over $G_p=121$ evaluation points captures this.

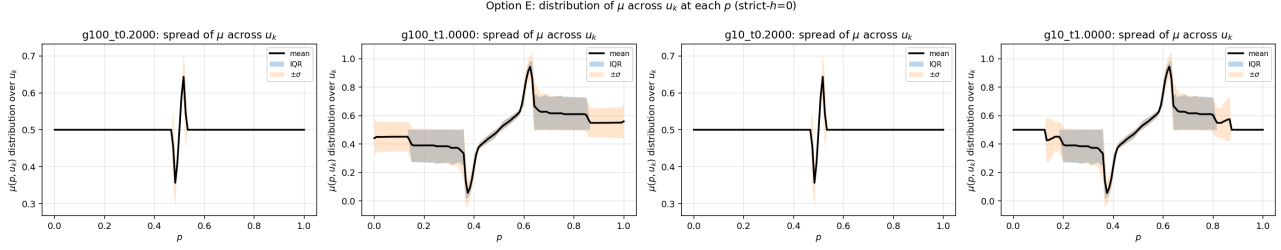
Why max is large near $p \in \{0, 1\}$. The cube cells P_{ij} cluster around the middle of $(0, 1)$ at high τ ; very few cells have P near the boundary. The PCHIP derivative in those sparse regions extrapolates badly. A fix: trim or replace the extreme p -values with the strict- $h=0$ result (cheap, accurate, no need to discard the bulk Option B estimate).

4 Lookup-curve overlays at sample cells



Top: $\mu(p, u_k)$ for several u_k , comparing strict- $h=0$ (black) to cube-KB at $h=0.2$ (blue dashed) and Option B (green dash-dotted). Option B matches strict- $h=0$ across the bulk of p ; cube-KB systematically shifts. **Bottom:** signed differences. The Option B difference is essentially zero everywhere except the extreme tails (where cube cells are sparse); cube-KB has a smooth, systematic bias.

5 Option E: distribution of μ across u_k at each p (diagnostic)



At each p , $\{\mu_{\text{strict}}(p, u_k)\}_k$ is a 1D distribution over the 11 signal nodes. The plots show the mean (black), interquartile range, and $\pm\sigma$ bands. This is a **diagnostic, not a lookup builder**: it shows where in p -space the price aggregates information most informatively.

Patterns visible.

- At $\tau=0.2$ (low signal precision): the distribution is tight around 0.5 across all p — the price is largely uninformative.
- At $\tau=1.0$: μ spreads widely across u_k at intermediate p — the price observation strongly modulates the posterior for agents with informative signals.

This is the *spread* of the lookup function across the agent dimension, the natural complement to its p -dependence.

6 Recommendations

1. **Adopt Option B (empirical CDF + PCHIP density) as the production lookup-table builder.** It is exact-in-median, has no bandwidth parameter, and runs in ~ 10 ms per cell — faster and more accurate than the existing GL kernel-band (~ 400 ms).
2. **Trim or fallback at the extremes.** In the $\sim 5\%$ of p -values near $\{0, 1\}$ where cube cells are sparse, replace the PCHIP derivative by a fallback (strict- $h=0$ direct, or kernel-band with a p -adaptive bandwidth). This bounds the max error.
3. **Re-do the $K=3$ (γ, τ) sweep with Option B.** The deficit-ridge result will (almost certainly) shift toward the strict- $h=0$ values we saw in the previous report: deficit ~ 0.006 at high τ instead of ~ 0.27 .
4. **Option C (MC Bayes) is useful as a third-party cross-check** but expensive — consider it a diagnostic, not a production builder.
5. **Option D (level-set MC) doubles as a sanity check on strict- $h=0$:** median ~ 0 at $\tau=0.2$ across all γ confirms strict is the right ground truth.

7 Reproducibility

- Code: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_lookup_alts.py` (the five builders) and `dd_k3_lookup_alts_run.py` (the runner).

- Per-cell μ -tables for all five options: `projects/REZN/solved_fixed_points/dd_k3_overnight/alts_npz/dd_k`
- Summary JSON: `.../dd_k3_overnight/alts_results.json`.